CS3001 – Computer Networks
Quiz – 3B

Name	
Roll. No.	
Section	

CLO1: Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Part 1 [7 Marks]:

In rdt3.0 the sender again uses sequence numbers {0, 1}. This time, the physical channel may corrupt packets but does not lose them, and the sender's timeout is set greater than RTT. However, the ACK for the first packet is repeatedly corrupted several times in a row before one finally arrives uncorrupted. Assume K retransmissions.

Starting from when the application hands the first data segment to the transport, describe:

1. **(In short bullet points, step-by-step)** Events, state transitions, and messages exchanged until the sender reaches "Wait for call 1 from above."
2. How many data transmissions of seq 0 occur, how many ACKs are sent by the receiver?
3. Why is the data still delivered exactly once at the receiver despite multiple retransmissions?

Solution:

1.

- Sender in "Wait for call 0 from above" sends pkt0, starts timer, enters "Wait for ACK0."
- Receiver gets the first arrival of pkt0 not corrupted; expected seq = 0, so it delivers the payload once and sends ACK0.
- Several ACK0 instances are corrupted in transit. Each corrupted ACK arrival leaves the sender in "Wait for ACK0"; no stop_timer; timer eventually expires.
- On each timeout, the sender retransmits the same pkt0, restarts the timer, stays in "Wait for ACK0."
- Receiver receives each retransmitted pkt0 (uncorrupted), recognises it as a duplicate (seq 0 already delivered), suppresses delivery/drops it, but re-sends ACK0.
- Eventually, an uncorrupted ACK0 arrives at the sender; sender stops the timer and transitions to "Wait for call 1 from above."

2. Data transmissions with seq 0: 1 initial + k retransmissions (one per timeout while ACKs were corrupted), total k+1.
3. The receiver's state machine discards duplicates after sequence-number comparison.

Part 2 [3 Marks]:

A Selective Repeat (SR) receiver has window size N = 4 and expects sequence number 5 next. It then receives packets with sequence numbers in this order: 7, 5, 6, 5.

1. Which of these packets does the receiver deliver to the application (and in what order)?
2. Which ACKs does the receiver send (list them in order sent)?
3. After processing all four arrivals, what is the receiver's new expected sequence number?

Solution:

1. Delivered (in order): 5, 6, 7
2. ACKs sent (in order): ACK7, ACK5, ACK6, ACK5
3. New expected sequence number (rcv_base): 8